Deep Learning for Image Classification and Retrieval

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***Abstract*—This report investigates deep learning approaches for image classification using CIFAR-10 and STL-10. A convolutional neural network (CNN) is designed and evaluated on CIFAR-10 through hyperparameter experiments involving learning rate, batch size, and dropout. On STL-10, ResNet-18 and MobileNetV3-Large are evaluated using randomly initialized and ImageNet-pretrained weights. A middle fusion approach is implemented by combining the feature representations of both pretrained models. Finally, image retrieval is evaluated using cosine similarity and embeddings optimized through Triplet Loss. The experiments show that ImageNet pretraining substantially improves STL-10 classification and that middle fusion provides the strongest classification result. In contrast, the tested Triplet Loss configuration reduces retrieval performance for both architectures.**

***Index Terms*—Deep Learning, CNN, CIFAR-10, STL-10, Image Classification, Transfer Learning, Feature Fusion, Metric Learning, Image Retrieval**

I. Part I: CNN Classification on CIFAR-10

A. CNN Architecture

A convolutional neural network (CNN) was implemented from scratch for image classification on the CIFAR-10 dataset. The architecture was designed to learn hierarchical visual representations while respecting the assignment constraint of using no more than six convolutional layers. Its structure is illustrated in Fig. 1.

The network comprises three convolutional blocks, each containing two convolutional layers with 3×3 kernels and a padding of one. The number of feature channels increases progressively from 64 to 128 and 256 across the three blocks. Each convolutional layer is followed by batch normalization and a ReLU activation. A 2×2 max-pooling operation is applied after each block, reducing the spatial resolution of the feature maps.

For an input image of size 32×32 with three colour channels, the spatial dimensions are successively reduced to 16×16, 8×8, and 4×4. Consequently, the final feature maps have dimensions $256 \times 4 \times 4$, resulting in a flattened representation of 4096 features.

The classification head consists of a fully connected hidden layer with 256 neurons, followed by a ReLU activation and a dropout layer. A final fully connected layer produces ten output scores, corresponding to the CIFAR-10 classes. The implemented network contains 2,198,602 trainable parameters and six convolutional layers.

This architecture combines progressive feature extraction with batch normalization and dropout. Increasing the number of channels allows the network to learn increasingly complex feature representations, while max-pooling reduces the spatial dimensions of the feature maps.

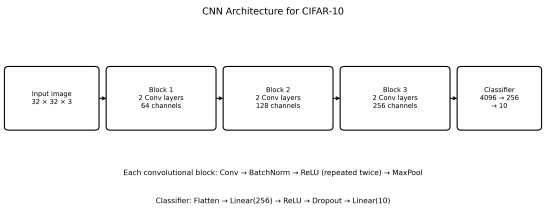


Fig. 1. Architecture of the six-convolutional-layer CNN implemented for CIFAR-10 classification.

B. Experimental Setup

The CIFAR-10 dataset contains 50,000 training images and 10,000 test images, distributed across ten classes. All images have a resolution of 32×32 pixels and three colour channels.

Before training, the images were normalized using the dataset-specific channel means and standard deviations. Data augmentation was applied exclusively to the training subset, consisting of random cropping with padding of four pixels and random horizontal flipping. The validation and test subsets were only normalized, without additional augmentation.

The original training set was randomly divided into training and validation subsets, containing 80% (40,000 images) and 20% (10,000 images) of the samples, respectively. A fixed random seed of 42 was used to ensure reproducibility of the split. The official CIFAR-10 test set was kept separate and used only for the final evaluation.

Four experiments were conducted to investigate the effect of different hyperparameter configurations on classification performance. The learning rate, batch size, and dropout probability were varied, while the remaining training settings were kept fixed.

All experiments used the AdamW optimizer with a weight decay of 10^{-4} and the cross-entropy loss function. Training was performed for 20 epochs, using a cosine annealing learning-rate scheduler.

Mean Class Accuracy (mCA) was used as the primary metric for selecting the best model configuration. It is defined as

$$\mathrm{mCA} = \frac{1}{C}\sum_{i=1}^{C}\frac{TP_i}{N_i}, \tag{1}$$

where C denotes the number of classes, TP_i is the number of correctly classified samples belonging to class i, and N_i is the total number of samples of that class.

Macro-averaged precision, recall, and F1-score were also computed to provide a complementary evaluation of classification performance.

Table I summarizes the hyperparameter configurations and their corresponding best validation mCA.

TABLE I
HYPERPARAMETER CONFIGURATIONS AND BEST VALIDATION MCA FOR CIFAR-10.

Experiment	Learning rate	Batch size	Dropout	Val. mCA (%)
EXP1	0.0010	128	0.3	87.59
EXP2	0.0005	128	0.3	**88.76**
EXP3	0.0010	64	0.3	88.31
EXP4	0.0010	128	0.5	87.38

C. Results and Discussion

Among the four evaluated configurations, EXP2 achieved the highest validation mCA, reaching 88.76% at epoch 19. Compared with the baseline configuration (EXP1), which used a learning rate of 0.001, EXP2 employed a lower learning rate of 0.0005. This resulted in an improvement of approximately 1.17 percentage points in validation mCA.

Reducing the batch size from 128 to 64 in EXP3 also improved performance, resulting in a validation mCA of 88.31%. In contrast, increasing the dropout probability from 0.3 to 0.5 in EXP4 resulted in a lower validation mCA of 87.38%. Therefore, neither modification improved the validation performance relative to the baseline.

Overall, the results suggest that the lower learning rate provided the most effective improvement among the configurations evaluated. Consequently, EXP2 was selected for the final evaluation on the official CIFAR-10 test set.

At its best validation epoch, EXP2 achieved a training loss of 0.2451 and a validation loss of 0.3429. These values indicate a difference between training and validation performance, although this difference alone is not sufficient to establish severe overfitting.

The selected model achieved an accuracy and mCA of 88.10% on the test set, exceeding the required 85% threshold. The similar values obtained for accuracy, macro-averaged recall, and F1-score indicate consistent classification performance across the evaluated metrics.

1) Learning Curve Analysis: Figure 2 presents the training and validation loss curves for the four hyperparameter configurations over the 20 training epochs.

TABLE II
TEST-SET PERFORMANCE OF THE SELECTED CNN CONFIGURATION (EXP2).

Metric	Test score (%)
Accuracy	88.10
mCA	88.10
Precision (macro)	88.14
Recall (macro)	88.10
F1-score (macro)	88.11

All experiments show a general decrease in cross-entropy loss throughout training, indicating that the models progressively learned useful representations from the training data. Although some fluctuations are observed in the validation curves during the initial epochs, the validation loss generally stabilizes as training progresses. No pronounced increase in validation loss is observed towards the end of training, suggesting no clear evidence of severe overfitting.

EXP2 consistently achieves comparatively low training and validation losses, with a validation loss of approximately 0.34 at its best validation epoch. This behaviour is consistent with its superior validation mCA.

EXP4 exhibits higher training loss than the other configurations, which is consistent with the stronger regularization introduced by its higher dropout probability. Its validation performance also remains below that of the baseline and EXP2.

Overall, the learning curves support the selection of EXP2, which combines the highest validation mCA with comparatively low training and validation losses.

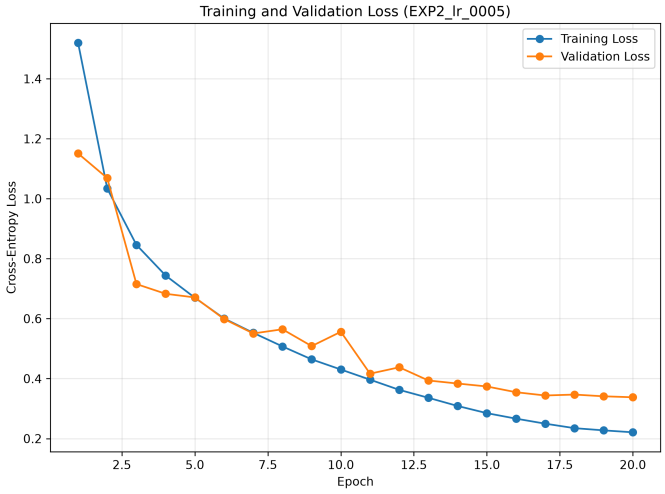


Fig. 2. Training and validation loss curves for the four CIFAR-10 experiments.

2) Confusion Matrix Analysis: The class-wise performance of the selected EXP2 model is examined through the confusion matrix shown in Fig. 3. The matrix provides insight into the distribution of correct predictions and misclassifications across the ten CIFAR-10 classes.

The observed results indicate stronger classification performance for vehicle-related classes, particularly automobiles, ships, trucks, and airplanes. In contrast, animal classes such as cats and dogs appear more challenging to classify.

The most noticeable confusion occurs between cats and dogs, with samples from each class frequently being assigned to the other. This may be partly explained by their similar visual characteristics, including body shape, fur, and facial features, which can be difficult to distinguish at the low image resolution of CIFAR-10.

Additional confusion is observed between automobiles and trucks, as well as between airplanes and ships. These errors may be related to similarities in object appearance, shape, or background context.

Overall, the confusion matrix suggests that, although the CNN achieves strong aggregate classification performance, its predictions remain more challenging for visually similar categories.

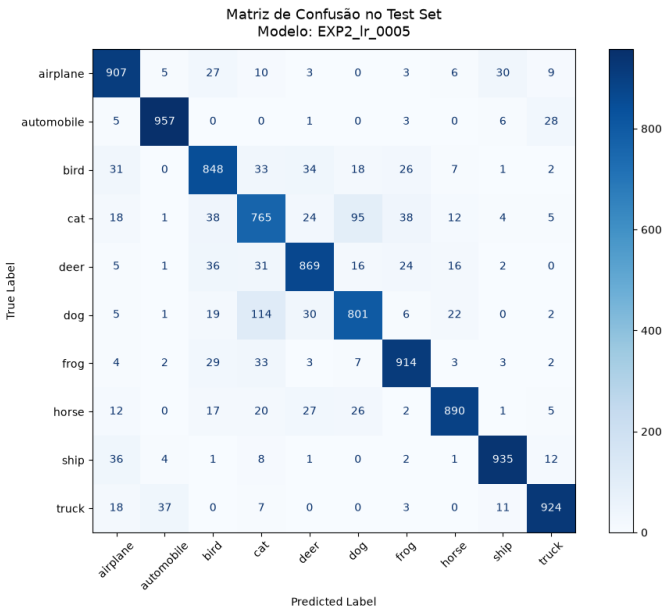


Fig. 3. Confusion matrix of the selected CNN (EXP2) on the CIFAR-10 test set.

In summary, the selected configuration achieved a test mCA of 88.10%, exceeding the required 85% threshold while respecting the maximum of six convolutional layers.

II. Part II: STL-10 Image Classification

A. Experimental Setup

The STL-10 benchmark was employed to evaluate two distinct convolutional architectures: ResNet-18 and MobileNetV3-Large. Each network was evaluated under two initialization protocols: training from scratch and utilizing ImageNet-pretrained weights.

The dataset consists of 10 classes, with 5,000 images in the official training partition and 8,000 images in the official test partition. To monitor training progress and evaluate generalization performance, the training set was partitioned into 4,000 training images and 1,000 validation images using a stratified 80/20 split with a fixed random seed. The official test partition remained isolated and was used strictly for the final benchmark evaluation.

Images were processed at their native resolution of 96×96 pixels. Data augmentation during training consisted of random crops with an 8-pixel padding and random horizontal flips with a probability of 0.5. Validation and test samples were evaluated deterministically without spatial transformations. All samples were normalized using standard ImageNet channel-wise mean and standard deviation vectors. The DataLoaders were configured with a batch size of 64.

The four individual configurations were optimized for 10 epochs using the AdamW optimizer with an initial learning rate of 10^{-3} and a weight decay of 10^{-4}, minimizing the standard cross-entropy objective.

For the models trained from scratch, all parameters were updated. For the pretrained models, the ImageNet feature extraction layers were frozen, retaining batch normalization layers in evaluation mode, and only the final linear classification heads were optimized. In all experiments, the model checkpoint corresponding to the highest validation accuracy was retained for final evaluation.

Final test performance was evaluated using overall accuracy, macro-averaged precision, macro-averaged recall, and macro-averaged F1-score. Training and validation loss curves and confusion matrices were collected to analyze learning dynamics and class-specific behaviours.

B. Classification Results and Discussion

Table III reports the test set performance for the four standalone model configurations.

TABLE III
STL-10 CLASSIFICATION RESULTS ON THE TEST SET.

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
ResNet-18 (Scratch)	58.79	58.33	58.79	57.20
MobileNetV3-Large (Scratch)	42.70	50.41	42.70	40.57
ResNet-18 (Pretrained)	86.89	87.07	86.89	86.85
MobileNetV3-Large (Pretrained)	88.84	88.89	88.84	88.82

The results show a substantial advantage for ImageNet pretraining in both architectures. ResNet-18 increased from 58.79% accuracy when trained from scratch to 86.89% with pretrained weights, a gain of 28.10 percentage points. MobileNetV3-Large increased from 42.70% to 88.84%, a gain of 46.14 percentage points. These differences indicate that, under the limited training set and ten-epoch budget, pretrained feature representations provide a much stronger starting point than random initialization.

Among the scratch models, ResNet-18 outperformed MobileNetV3-Large by 16.09 percentage points. The ranking was reversed when pretrained weights were used: MobileNetV3-Large achieved 88.84%, compared with 86.89%

for ResNet-18. This reversal shows that architecture performance depends on the initialization and training regime, rather than on the architecture alone.

The comparison involves different optimization settings. All parameters were updated in the scratch configurations, whereas only the final classification layer was trained in the pretrained configurations. Consequently, the measured gains reflect both the use of ImageNet initialization and the corresponding frozen-backbone strategy.

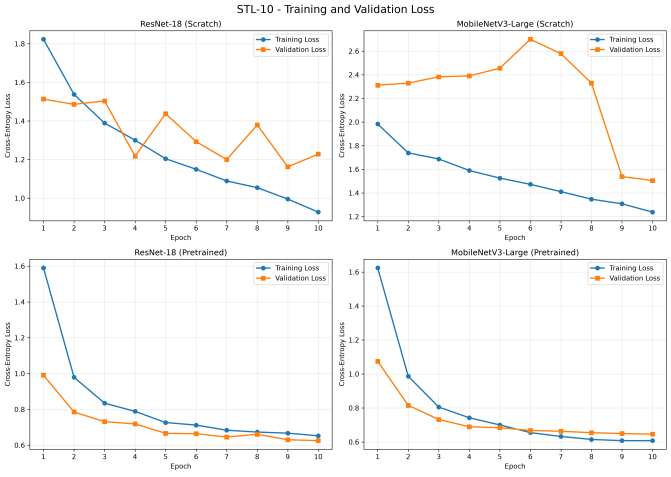


Fig. 4. Training and validation loss curves for the four standalone STL-10 models and the middle-fusion model.

The loss trajectories in Fig. 4 illustrate the optimization behaviour of all five configurations. The pretrained models converge to substantially better test performance than the scratch models, consistent with the benefit of ImageNet representations under the ten-epoch training budget. The middle-fusion trajectory is also included and is discussed separately below.

Inspection of the confusion matrices in Fig. 5 reveals differences in class-specific classification performance. The models trained from scratch exhibit a larger number of misclassifications, whereas the pretrained models show stronger concentrations along the main diagonal.

For the pretrained ResNet-18, the strongest class-wise F1-scores were obtained for *car* (0.92) and *bird* (0.91), while *cat* and *dog* were the most challenging classes (0.80 each). MobileNetV3-Large achieved particularly strong results for *car* and *ship* (0.94 each), with a more balanced distribution across the remaining categories. These class-level results complement the macro-averaged metrics and show that the architectures do not make identical class-specific errors.

C. Middle Fusion

To evaluate whether combining representations from both pretrained models provides performance improvements over individual baselines, a middle fusion model was implemented using ResNet-18 and MobileNetV3-Large.

Both pretrained feature extractors were kept frozen during fusion training. Their final classification layers were replaced with identity mappings. ResNet-18 outputs a 512-dimensional vector and MobileNetV3-Large produces a 960-dimensional vector after global average pooling. Concatenating both vectors results in a 1,472-dimensional feature vector per sample.

The concatenated representation was forwarded to a dedicated classification head consisting of a fully connected layer (1,472 to 512 units), a ReLU activation, dropout with $p = 0.3$, and a final linear layer mapping to the 10 STL-10 classes.

Only the fusion classification head was optimized, while the underlying backbones remained in evaluation mode. The fusion network was trained for 15 epochs using AdamW (learning rate of 10^{-3}, weight decay of 10^{-4}, batch size of 64) with cross-entropy loss. As with the individual configurations, the checkpoint corresponding to the highest validation accuracy was selected for test evaluation.

The fusion model attained a test accuracy of 90.61%, with a macro precision of 90.84%, a macro recall of 90.61%, and a macro F1-score of 90.59%. Table IV compares these results with the standalone pretrained models.

TABLE IV
COMPARISON OF INDIVIDUAL PRETRAINED MODELS AND THE MIDDLE FUSION APPROACH ON THE STL-10 TEST SET.

Model	Acc. (%)	Prec. (%)	Rec. (%)	F1 (%)
ResNet-18 (Pretrained)	86.89	87.07	86.89	86.85
MobileNetV3-Large (Pretrained)	88.84	88.89	88.84	88.82
Middle Fusion	**90.61**	**90.84**	**90.61**	**90.59**

Middle fusion outperformed both individual pretrained models, with an accuracy gain of 3.72 percentage points over ResNet-18 and 1.77 percentage points over MobileNetV3-Large. Its best validation accuracy was 92.10% at epoch 11, while test accuracy reached 90.61%. The training loss decreased throughout the 15 epochs, whereas validation loss fluctuated and did not decrease monotonically. The difference between the best validation and test accuracies is limited, although the validation trajectory suggests that further training did not produce consistent improvements in generalization.

The improvement achieved by middle fusion suggests that the feature representations extracted by ResNet-18 and MobileNetV3-Large provide complementary information for the STL-10 classification task. The fusion model also achieved its strongest class-wise F1-scores for *ship* (0.95), *bird* (0.94), and *car* (0.93), while *cat* remained comparatively challenging (0.84). Although both backbones were pretrained on ImageNet, their distinct architectures may capture different visual attributes. By concatenating their feature vectors, the classification head can exploit information from both extractors, resulting in higher classification performance than either individual model.

Overall, middle fusion achieved the highest test performance among the evaluated approaches, highlighting the practical benefit of combining pretrained representations for image classification.

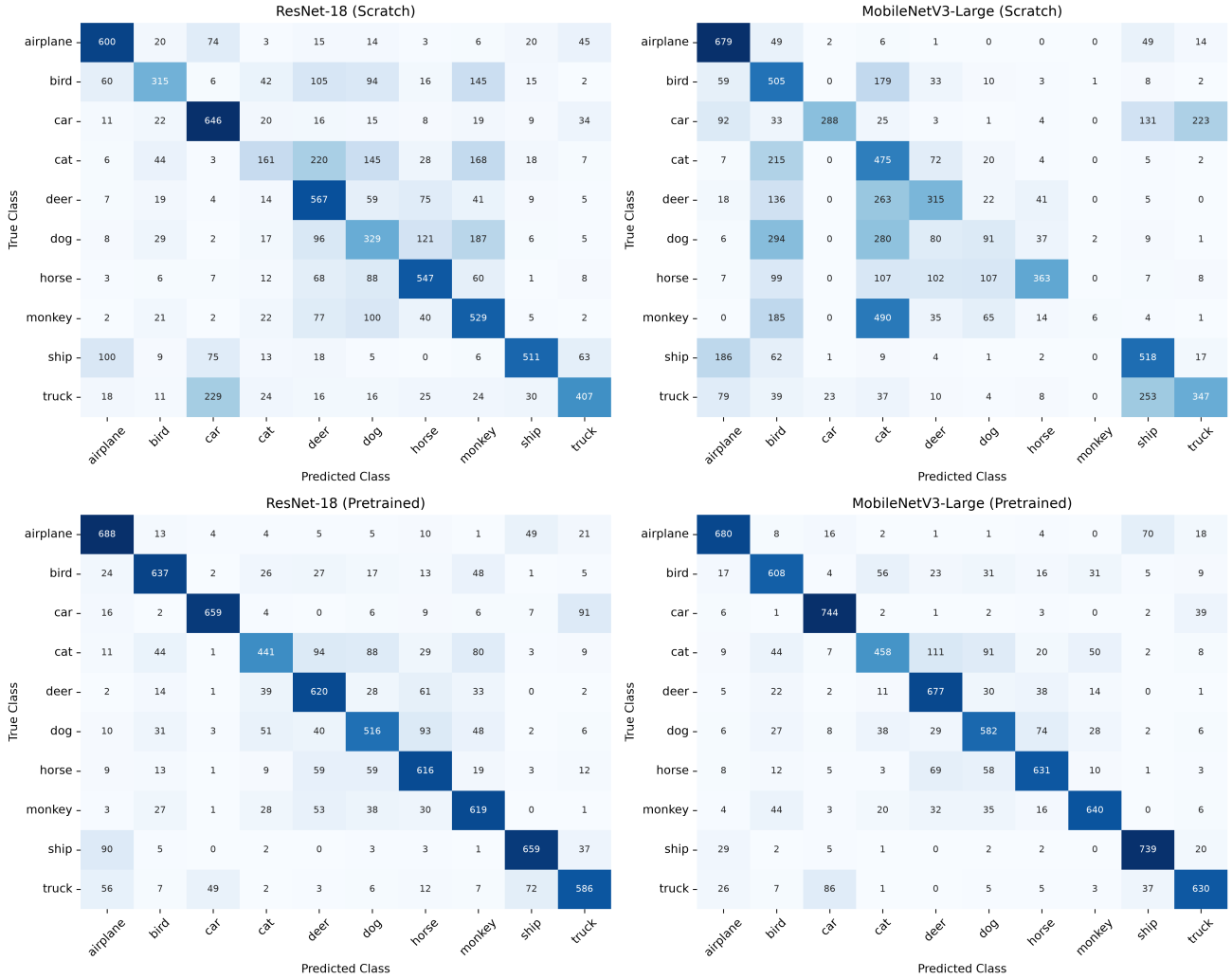


Fig. 5. STL-10 test confusion matrices for the four standalone configurations and the middle-fusion model.

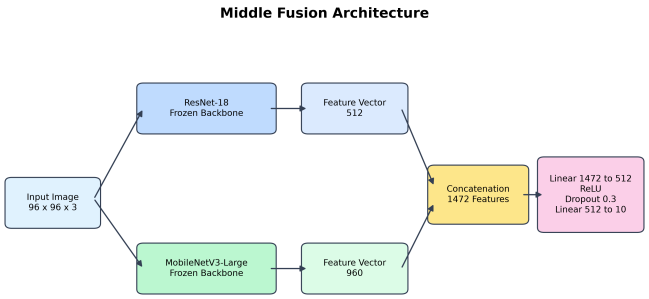


Fig. 6. Middle fusion architecture combining the feature representations of ResNet-18 and MobileNetV3-Large.

III. Part III: Image Retrieval

A. Experimental Setup

The image retrieval experiments were designed to evaluate whether visual representations can rank gallery images according to their semantic similarity to a given query image, without relying on class predictions.

The official STL-10 test partition (8,000 images) was divided into a query set containing 1,600 images (20%) and a gallery set containing 6,400 images (80%). This division was carried out using stratified sampling with a fixed random seed independently across each of the 10 classes (yielding exactly 160 query and 640 gallery images per class), guaranteeing disjoint subsets ($Q \cap G = \emptyset$). Metric-learning optimization was performed strictly on the training partition to prevent data leakage.

For each query, gallery samples were ranked by descending cosine similarity. Feature vectors were normalized to unit Euclidean length (ℓ_2 norm) prior to evaluation, allowing cosine similarities to be computed directly via matrix multiplication ($S = QG^\top$).

Embeddings were extracted from the pretrained classification backbones after removing their classification layers. For ResNet-18, the removal of the final fully connected layer yielded 512-dimensional descriptors. For MobileNetV3-Large,

replacing the final linear classifier with an identity mapping preserved the intermediate projection layer, yielding 1,280-dimensional descriptors.

To investigate metric learning, both backbones were further trained for 5 epochs using a batch-hard Triplet Loss with a margin of $m = 0.2$. Optimization employed AdamW with a learning rate of 10^{-5} and a weight decay of 10^{-4} on batches of size 64. Feature vectors were normalized to unit length before computing pairwise distances. Batch-hard mining selects, for each anchor embedding a, the most distant positive p of the same class and the closest negative n of a different class within the mini-batch:

$$\mathcal{L}_{\text{triplet}} = \frac{1}{|V|} \sum_{i \in V} \max\left(0,\, d(a_i, p_i) - d(a_i, n_i) + m\right), \quad (2)$$

where $d(\cdot, \cdot)$ denotes the Euclidean distance over normalized vectors and V is the set of anchors containing valid positive and negative counterparts within the batch. Checkpoints were retained based on the lowest validation loss.

Retrieval effectiveness was assessed via Precision at 1 (P@1), Precision at 5 (P@5), and mean Average Precision (mAP). Precision at cutoff K is defined as:

$$P@K = \frac{1}{K} \sum_{k=1}^{K} rel(k), \quad (3)$$

where $rel(k) \in \{0, 1\}$ indicates whether the gallery sample at rank k matches the query class. Average Precision (AP) for a query is given by:

$$AP = \frac{1}{N_{\text{rel}}} \sum_{k=1}^{N_g} P(k) \cdot rel(k), \quad (4)$$

where $P(k)$ denotes the precision computed over the first k retrieved gallery samples, $N_{\text{rel}} = 640$ is the total number of relevant images in the gallery for that class, and $N_g = 6,400$ is the total gallery size. Averaging across all 1,600 queries yields the mAP.

B. Quantitative Results and Discussion

Table V reports the retrieval performance across both architectures, before and after fine-tuning with Triplet Loss.

TABLE V
IMAGE RETRIEVAL PERFORMANCE ON THE STL-10 TEST SET.

Model	P@1 (%)	P@5 (%)	mAP (%)
ResNet-18 (Original)	85.25	**81.69**	**50.63**
ResNet-18 (Triplet Loss)	74.31	71.37	43.14
MobileNetV3-Large (Original)	**85.62**	81.65	44.24
MobileNetV3-Large (Triplet Loss)	74.31	69.89	32.46

The original representations extracted from the classification baselines achieved high Top-K precision. MobileNetV3-Large yielded the highest P@1 (85.62%), while ResNet-18 achieved the highest P@5 (81.69%) and the highest overall mAP (50.63%). The gap between high P@1/P@5 and moderate mAP observed in both models reflects the retrieval scenario: while the nearest neighbours in the embedding space frequently belong to the query category, relevant samples located further down the 6,400-image gallery become interspersed with other classes.

Training with batch-hard Triplet Loss resulted in a reduction across all evaluation metrics for both models. For ResNet-18, P@1 decreased from 85.25% to 74.31%, P@5 dropped from 81.69% to 71.37%, and mAP declined by 7.49 percentage points (from 50.63% to 43.14%). Similarly, MobileNetV3-Large exhibited reductions across all metrics, with mAP falling from 44.24% to 32.46%.

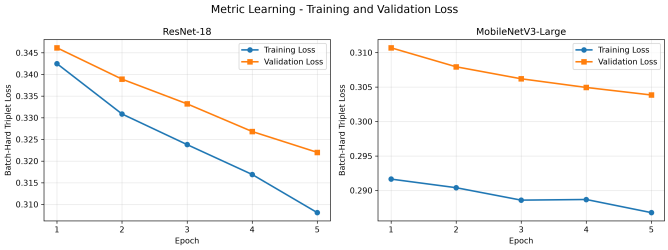


Fig. 7. Training and validation batch-hard Triplet Loss trajectories for ResNet-18 and MobileNetV3-Large over five epochs.

The loss trajectories shown in Fig. 7 indicate a gradual decrease in both training and validation losses. For ResNet-18, training loss decreased from 0.3421 to 0.3085, while validation loss decreased from 0.3480 to 0.3220. For MobileNetV3-Large, training loss decreased from 0.2923 to 0.2856 and validation loss from 0.3208 to 0.3015.

One possible explanation for the performance degradation is the use of randomly sampled mini-batches instead of class-balanced sampling strategies, such as $P \times K$ sampling. Although batch-hard mining selects the hardest positive and negative examples available within each batch, the quality of these selections depends on the class composition of the mini-batch. Furthermore, fine-tuning the entire backbone may have altered useful representations acquired during ImageNet pre-training. These factors may have contributed to the observed reduction in retrieval performance.

C. Qualitative Evaluation

Qualitative inspection provides insight into how the embedding space structures visual similarities. Across the 1,600 test queries evaluated with the ResNet-18 Triplet Loss model, 754 queries (47.13%) attained perfect Top-5 precision (5/5 correct), whereas 120 queries (7.50%) retrieved no relevant images within the top five results.

In successful cases, queries typically present clear object contours and distinctive shapes that facilitate class separation. In failure cases, retrieved gallery images often share global visual characteristics with the query, such as dominant background colors, textures, or broad animal silhouettes, even when they belong to different categories.

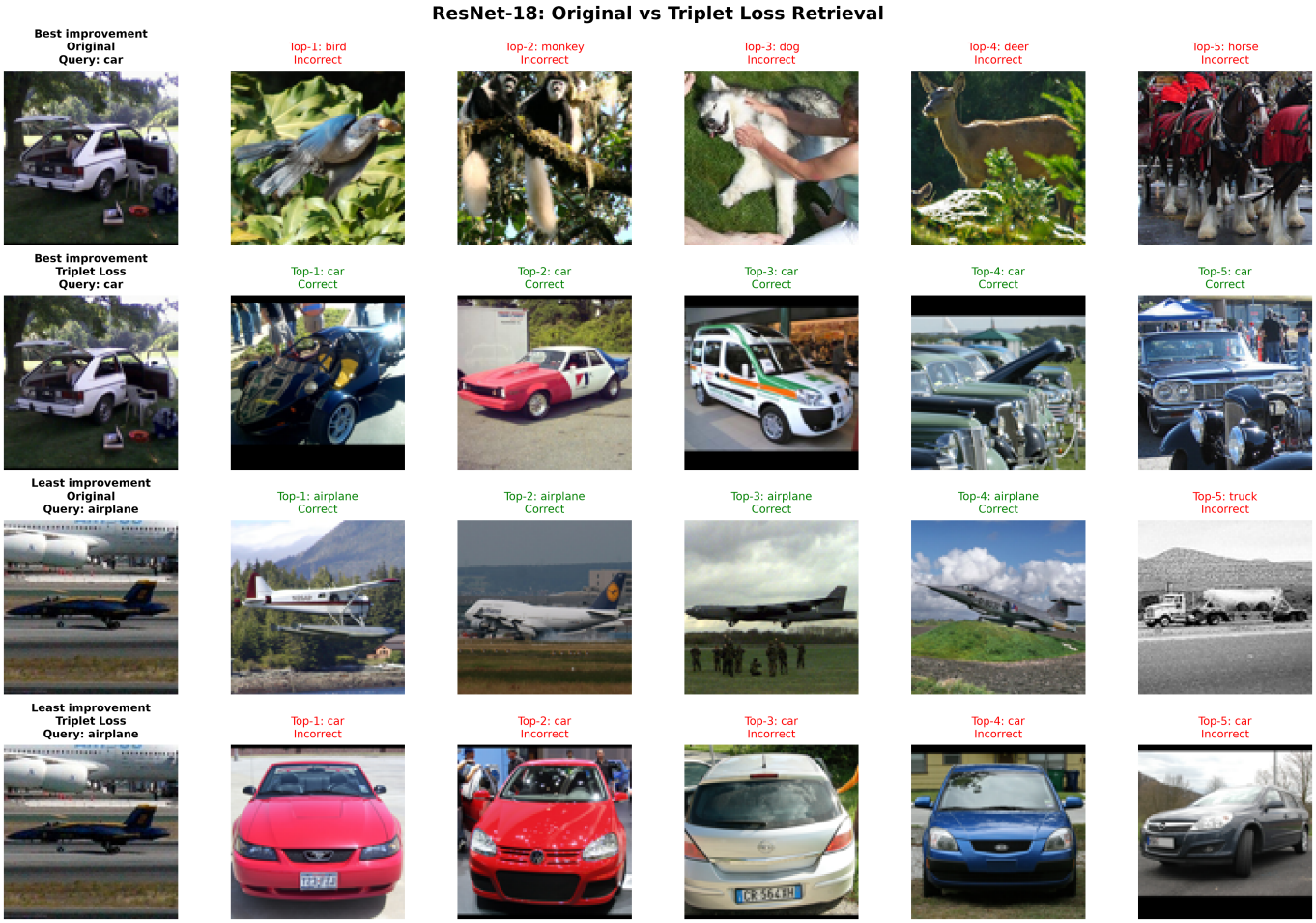


Fig. 8. Qualitative comparison of ResNet-18 retrieval before and after Triplet Loss optimization across selected queries. The same query images are used in both configurations to ensure a direct comparison. Green headings denote correct class matches; red headings indicate mismatches.

As shown in the comparative analysis in Fig. 8, metric learning alters ranking trajectories locally: certain queries exhibited improved Top-5 rankings after fine-tuning (e.g., improving from 1/5 to 5/5 correct matches), whereas other queries experienced degradation (e.g., decreasing from 5/5 to 0/5 correct matches), reflecting the overall decrease in quantitative performance observed across the gallery.

D. Summary and Conclusion

The retrieval experiments demonstrate that classification accuracy does not directly determine ranking quality across an extensive gallery. Original pretrained embeddings without additional metric learning provided the most effective representations in this setup, with MobileNetV3-Large achieving 85.62% P@1 and ResNet-18 achieving 50.63% mAP.

These results suggest that fine-tuning deep representations with ranking objectives is sensitive to the sampling strategy and training configuration. In this experiment, the adopted batch-hard Triplet Loss setup did not improve the pretrained representations, highlighting the importance of investigating class-balanced sampling and alternative metric-learning configurations.

IV. Conclusion

This work investigated deep learning approaches for image classification and retrieval using the CIFAR-10 and STL-10 datasets. The experiments explored the impact of network architecture, hyperparameter configurations, transfer learning, feature fusion, and metric learning on visual representation performance.

For CIFAR-10, a custom convolutional neural network comprising six convolutional layers achieved a test accuracy and mean Class Accuracy (mCA) of 88.10%. The hyperparameter experiments highlighted the influence of learning rate, batch size, and dropout on model performance. Among the tested configurations, the model trained with a learning rate of 5×10^{-4}, a batch size of 128, and a dropout rate of 0.3 achieved the highest validation mCA. These results demonstrate that a relatively compact CNN can learn effective visual representations for image classification, while also highlighting the importance of hyperparameter selection.

For STL-10, experiments with ResNet-18 and MobileNetV3-Large showed the benefits of ImageNet pretraining. Both pretrained architectures substantially

outperformed their counterparts trained from scratch, achieving test accuracies of 86.89% and 88.84%, respectively. Furthermore, combining their feature representations through middle fusion improved classification performance, reaching 90.61% test accuracy and a macro F1-score of 90.59%. This result suggests that the complementary information captured by different network architectures can contribute to more effective representations when combined through feature-level fusion.

The image retrieval experiments further demonstrated that classification performance does not necessarily translate directly into retrieval effectiveness. Among the original pretrained representations, MobileNetV3-Large achieved the highest Precision at 1 (85.62%), whereas ResNet-18 obtained the highest Precision at 5 (81.69%) and mean Average Precision (mAP) (50.63%). Applying batch-hard Triplet Loss reduced all retrieval metrics for both architectures under the tested training configuration. This suggests that metric-learning performance is sensitive to factors such as mini-batch composition, sampling strategies, and fine-tuning procedures. Consequently, improving retrieval representations may require more carefully designed training and sample-mining strategies.

Overall, the experiments highlight that model performance depends not only on network architecture, but also on the training strategy and the specific objectives of each task. Transfer learning and feature-level fusion proved particularly effective for STL-10 classification, while the original pretrained embeddings provided stronger retrieval performance than the fine-tuned representations in the evaluated setup. These findings emphasize the importance of selecting and adapting visual representations according to the requirements of each image understanding task.